

Our land. Our energy future.

Investigating energy resilience through electricity, seasonal resources, land constraints, industrial systems and open modelling.

THE RESEARCH QUESTION

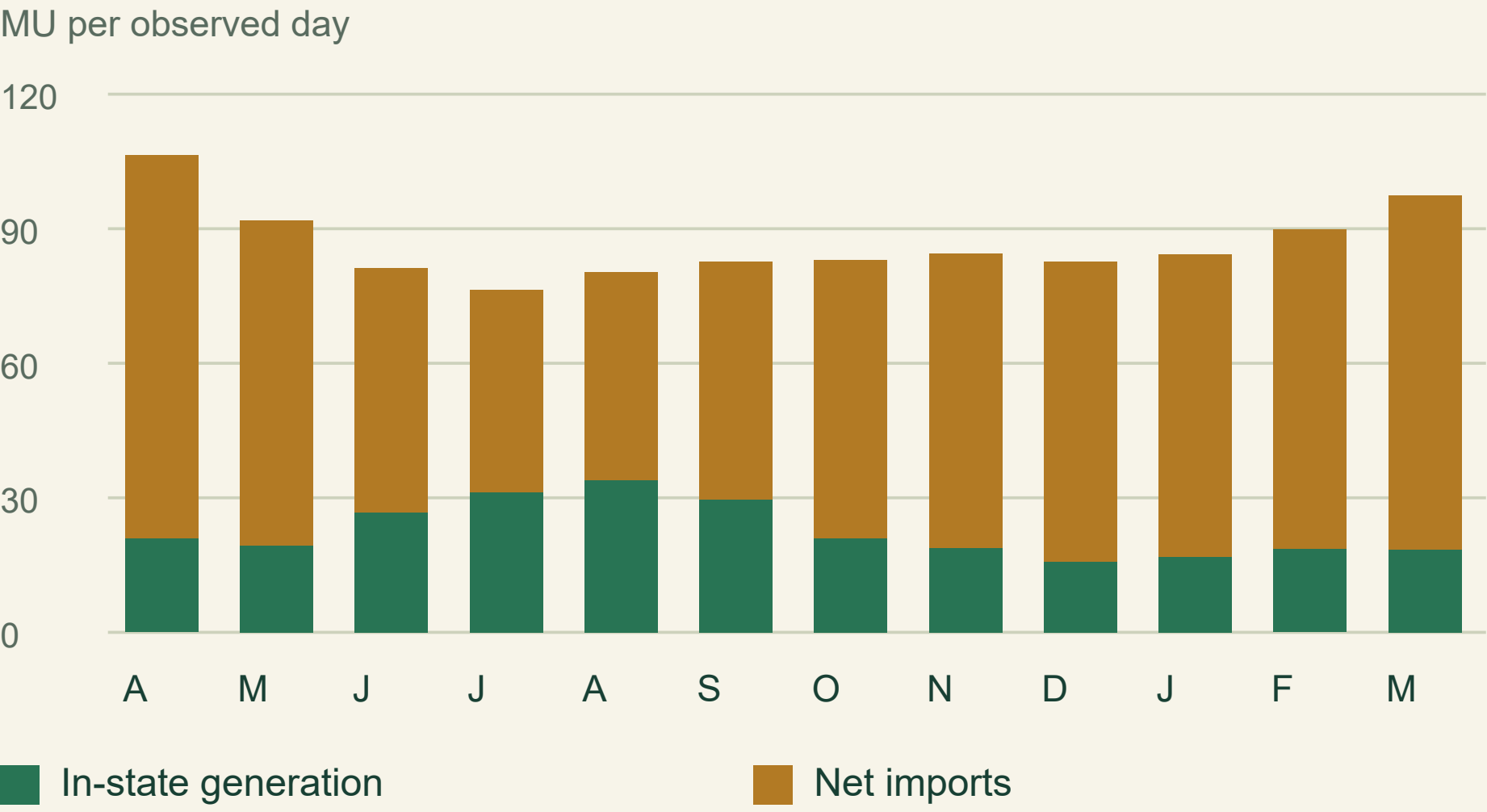
How can Kerala meet future demand reliably while respecting its land and water?

Early experiments show why timing and transfer capability matter alongside new generation. The current evidence supports comparison, not a final 2040 capacity plan.

01 / OBSERVED ELECTRICITY

A connected, seasonal system

73.8% of recorded consumption came from net imports.

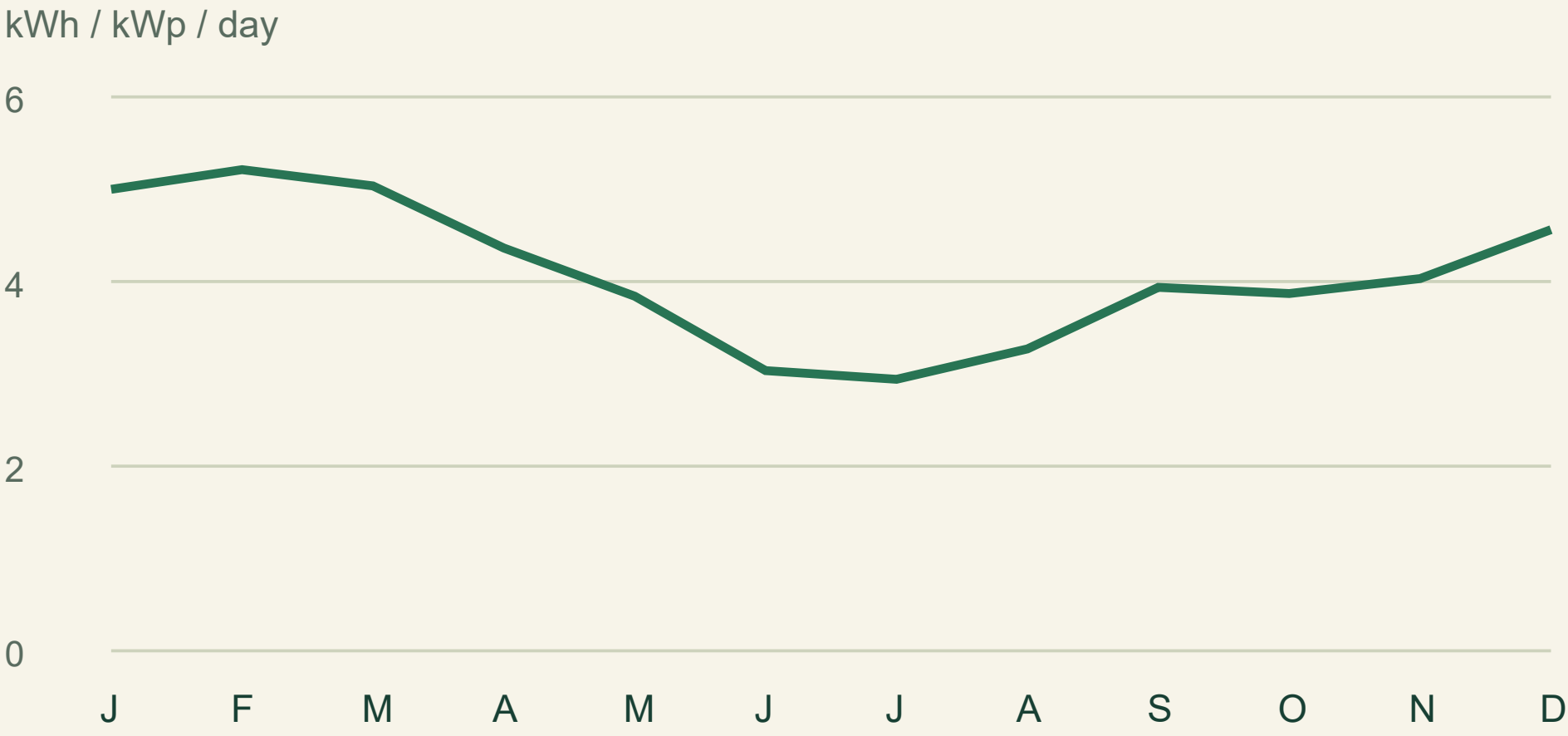


SLDC FY2024-25: 354 of 365 daily reports. The 11 missing days are not filled. Bars show available-day means, not complete monthly totals.

02 / RESOURCE & LAND

Solar follows the seasons

Monthly median source-grid PV output



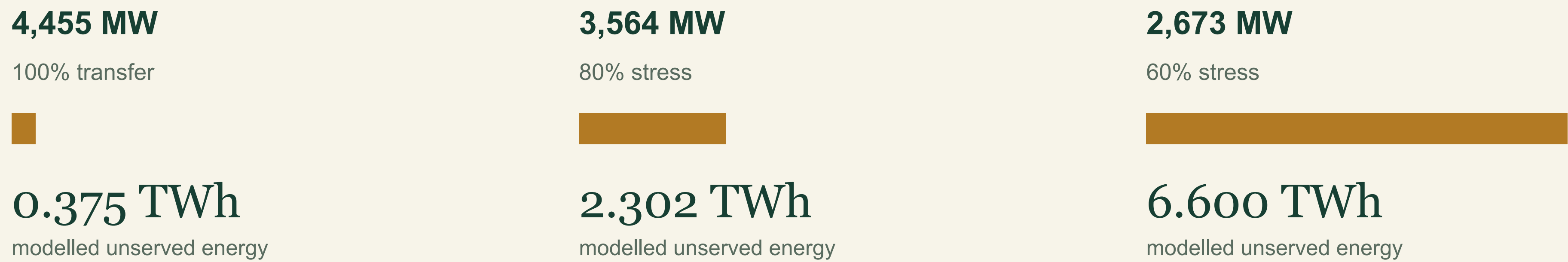
Resource is not permission to build.

Global Solar Atlas 2, 1999-2018 climatology. Wind, terrain, ecology and grid access must also be checked. Eligible land and project capacity are unresolved.

03 / 2030 MODEL EXPERIMENT

Transfer stress exposes a flexibility problem.

108 full-year economic sensitivities now combine demand, renewable envelopes, battery costs, transfer limits and import-price assumptions in PyPSA.



Selected case: reference FY2030-31 demand, reference renewable envelope, low battery cost. All candidate capacity limits bind. These shortages describe the assumptions of this model; they are not a forecast of actual power cuts.

METHOD

Preserve source records and gaps. Characterise resources. Test constrained dispatch and investment. Check numerical equivalence. Keep assumptions traceable.

WHAT STILL NEEDS WORK

Measured hourly demand; reservoir and cascade operation; outages; landed prices; statutory siting; project finance and grid feasibility.